

Zachariah Risheq

Chantilly, VA | +1 (804) 460-9237 | Zachkr05@gmail.com | linkedin.com/in/zachariah-risheq | github.com/zachkr05 | zachkr05.github.io

EDUCATION

University of Virginia — GPA: 4.00/4.00 <i>Master of Science in Computer Science, Advisor: Prof. Yen-Ling Kuo</i>	Expected May 2028 Charlottesville, VA
University of Virginia — Major GPA: 3.72/4.00 <i>Bachelor of Science in Computer Science, School of Engineering</i>	May 2026 Charlottesville, VA

EXPERIENCE

Air Force Research Laboratory (AFRL) <i>Research Intern (Advisor: Dr. Oliver Nina) — C++, Python</i>	May 2026 – August 2026 Remote
– Authored the research proposal (on Multi-Agent Reinforcement Learning and Diffusion Models) to investigate diffusion models for uncertainty-aware multi-agent reinforcement learning. – Python/PyTorch: Proposed and Built RAZR, a novel decentralized cooperative MARL system coupling MAPPO with a diffusion-based uncertainty signal, trained CTDE in an asymmetric actor-critic framework – Python/PyTorch: Built Pivot a low-dimensional test environment for a collaborative framework – Python/PyTorch: RAZR achieved an increase of +5% value estimation compared to baseline and a +8% increase success episodes per run compared to baseline PPO policy	
University of Virginia – LIVE Lab <i>Research Assistant (Advisor: Prof. Yen-Ling Kuo) — C++, Python, LLMs, ROS2</i>	Mar. 2026 – Present Charlottesville, VA
– Python/PyTorch/C++: Built an end-to-end text-to-MuJoCo simulation generation pipeline, synthesizing and reconstructing 3D assets from natural-language scene specifications. – Python/C++: Proposed and implemented novel Markov-Chain-Monte-Carlo (MCMC) simulation generation process utilizing value function for MDP – C#/Unity: Built a 100 Hz AR teleoperation system on Meta Quest 3, streaming bimanual controller poses to a mobile manipulator over low-latency sockets for scalable robot-demonstration collection. – Evaluated generated scenes across 100 prompts spanning 25 object classes, achieving 97% scene success – Drake/MuJoCo: Built a cross-format scene-conversion pipeline exporting model directives to MuJoCo and Unity, plus a post-hoc critic-driven evaluation stage over frozen 3D scenes	
University of Virginia – AMR Lab <i>Research Assistant (Advisor: Prof. Nicola Bezzo) — C++, C#, Python, LLMs, ROS1</i>	Aug. 2024 – June 2026 Charlottesville, VA
– C++17/Python/C#: Led a cross-functional team developing an interactive AR interface in Unity for real-time UAV/UGV trajectory visualization and mission-plan editing – Python/C++17: Developed an LLM/VLM command-grounding service mapping ambiguous natural-language instructions into structured robot trajectory edits and goal specifications – Python, PyTorch: Developed a conditional diffusion model for robot costmap and trajectory generation, using value-function-derived supervision from human trajectory edits to adapt future plans. – C++/Python: Built a real-time TCP communication server (100HZ communication) with length-prefixed framing, integrating a diffusion-model backend, Unity AR client, Vicon motion capture, and a ROSbot (UGV robot) into one closed-loop pipeline – Evaluated command grounding on 100 instructions, achieving 97.3% execution accuracy / reducing invalid plans by 99%	

RESEARCH PUBLICATIONS & THESIS

- Z. Risheq, N. Bezzo, E. Travis.** “GuidAR: Meta-Learned MoE Diffusion Costmaps From AR Trajectory Edits” 2026
– Undergraduate Senior Thesis, University of Virginia. [UVA Libra](#)

TECHNICAL SKILLS

Languages: C++17/20, Python, Java, C#, JavaScript, SQL, Bash, HTML/CSS
Robotics/ML: ROS 2, ROS 1, PyTorch, MuJoCo, NumPy, OpenCV, Eigen, scikit-learn, Navigation2, Flask, Django, React, Unity, Tailwind, STL, Boost (Asio/Beast) **Tools:** Linux / Unix, Multithreading, JSON/XML, OpenAI API, MongoDB Atlas, Gemini AI, Docker